

Project Report

Breast-Ultrasound-Classification

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[Deep Learning for Medical Imaging]

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1. INTRODUCTION

Breast cancer is one of the most common diseases affecting women worldwide. Early detection of tumors is important for proper treatment and improved survival rates. Ultrasound imaging is widely used for detecting breast abnormalities because it is safe and cost-effective. In this project, a Hybrid CNN–Quantum model is used to classify breast ultrasound images into benign, malignant, and normal categories, helping in automatic and efficient tumor detection.

2. DATASET DETAILS

The dataset used in this project is BUSI (Breast Ultrasound Images Dataset) stored in the folder: Dataset_BUSI_with_GT.

This dataset contains breast ultrasound images used to detect and classify tumors. The images belong to three different classes:

- Benign
- Malignant
- Normal

These images are used to train a deep learning model that can automatically classify breast tumors from ultrasound scans. The images are organized in folders according to their class labels, which allows them to be easily loaded using the 'ImageFolder' dataset loader from PyTorch.

3. METHODOLOGY

a) Data Handling and Preprocessing

Several preprocessing steps are applied to prepare the dataset for training.

I. Image Resizing:

All images are resized to: 128×128 pixels. This ensures that all images have the same dimensions before being fed into the neural network.

II. Data Augmentation:

To improve model generalization, data augmentation is used. The augmentation techniques include: Random Horizontal Flip and Image Normalization. These transformations help the model learn better features and reduce overfitting.

III. Normalization:

The images are normalized using the following mean and standard deviation values:

Mean = [0.485, 0.456, 0.406]

Std = [0.229, 0.224, 0.225]

Normalization improves training stability and helps the neural network converge faster.

b) Data Loading

The data is loaded using PyTorch DataLoader, which helps in efficient training.

Batch size used: Batch Size = 16

Two data loaders are created:

- Train Loader – loads training data
- Test Loader – loads testing data

Data loaders help in:

- Loading data in batches
- Shuffling training data
- Faster training

c) Handling Class Imbalance

The dataset may contain unequal numbers of images in each class. To solve this problem, class weights are calculated using: `compute_class_weight()` function. These weights are applied in the `CrossEntropyLoss` function, which helps the model treat minority classes fairly during training.

d) Dataset Splitting

The dataset is divided into three parts:

DATA TYPE	PERCENTAGE	PURPOSE
Training Data	70%	Used to train the model
Validation Data	15%	Used to tune the model
Test Data	15%	Used to evaluate final performance

This split ensures that the model is tested on unseen data to measure its real performance.

e) Model Training

TECHNIQUES USED IN THE PROJECT:

This project uses a Hybrid Quantum-Classical Deep Learning Model.

The main techniques used are:

I. Classical Convolutional Neural Network (CNN)

A CNN is used to extract important features from the ultrasound images. The CNN architecture includes:

- Convolution layers
- Batch normalization
- Activation functions
- Feature extraction layers

CNN helps the model detect patterns such as:

- Tumor shapes
- Texture patterns
- Abnormal regions

II. Quantum Neural Network Layer

A quantum circuit is added after the CNN layer using PennyLane.

Quantum computing concepts used include:

- Angle Embedding
- Quantum Gates
- Quantum Measurements

The quantum layer processes features extracted by the CNN and helps improve classification performance.

III. Hybrid QCNN Model

The final model is called HybridQCNN, which combines:

- Classical CNN
- Quantum neural network layer

Workflow of the model:

Input Image → CNN Feature Extraction → Quantum Circuit Layer → Final Classification

This hybrid approach attempts to combine the strengths of classical deep learning and quantum computing.

Training Process

The model is trained using the following settings:

PARAMETER	VALUE
Optimizer	Adam
Learning Rate	0.0005
Loss Function	CrossEntropyLoss
Epochs	40
Batch Size	16

The Adam optimizer is used to update model weights during training. The CrossEntropyLoss function is used because this is a multi-class classification problem.

4. EVALUATION METRICS

The model performance is evaluated using classification metrics from Scikit-learn.

The evaluation process collects:

- True labels
- Predicted labels

These are used to compute the following metrics:

a) Confusion Matrix

Shows how many images were correctly and incorrectly classified for each class.

b) Classification Report

Includes:

- Accuracy
- Precision
- Recall
- F1-score

These metrics help understand how well the model performs on each tumor category.

5. RESULTS

After training the Hybrid CNN–Quantum model for 40 epochs, the model was evaluated using the test dataset.

During training, the loss value gradually decreased, showing that the model was learning useful patterns from the ultrasound images.

Example training loss values:

Epoch 1 : 1.1408

Epoch 5 : 1.0559

Epoch 10 : 1.0360

Epoch 15 : 1.0126

Epoch 18 : 0.9922

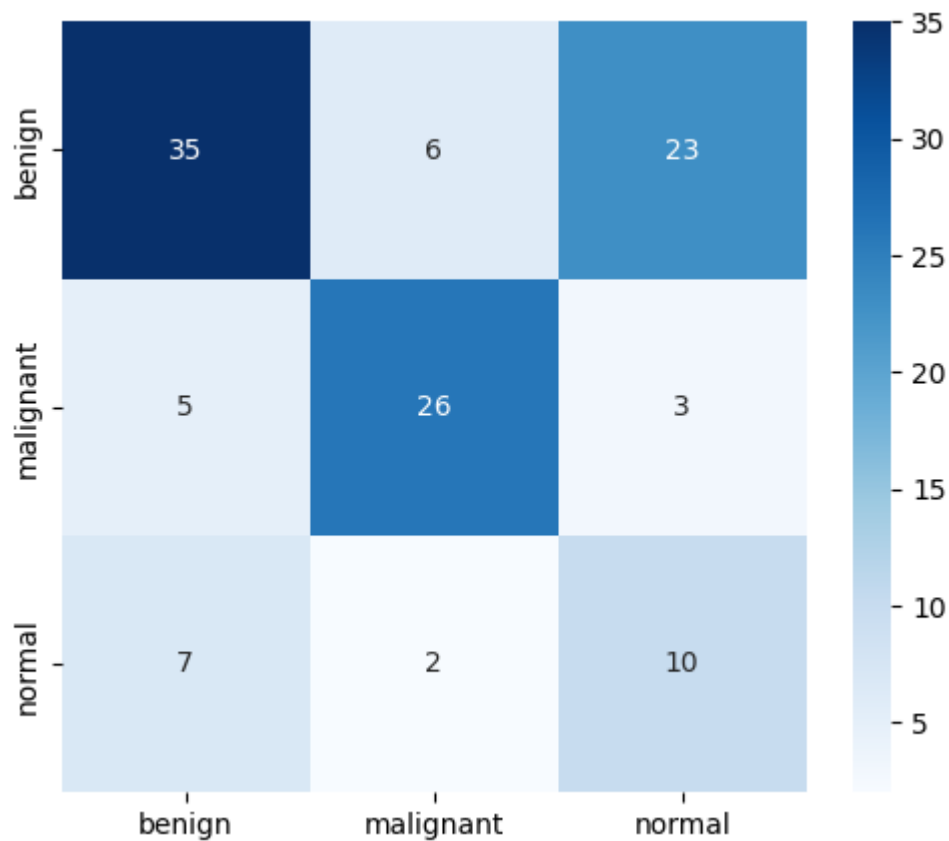
a) Classification Results

Class	Precision	Recall	F1-score	Support
Benign	0.74	0.55	0.63	64
Malignant	0.76	0.76	0.76	34
Normal	0.28	0.53	0.36	19

Overall Accuracy: 61%

Total Test Samples: 117 images

	precision	recall	f1-score	support
benign	0.74	0.55	0.63	64
malignant	0.76	0.76	0.76	34
normal	0.28	0.53	0.36	19
accuracy			0.61	117
macro avg	0.60	0.61	0.59	117
weighted avg	0.67	0.61	0.63	117



b) Interpretation of Results

From the evaluation results:

- The model performs best for malignant tumors, achieving an F1-score of 0.76.
- The benign class also shows reasonably good performance.
- The normal class has lower accuracy, likely due to fewer training samples and class imbalance.

The overall accuracy of 61% indicates that the hybrid CNN–Quantum model is able to learn meaningful features from ultrasound images but still has scope for improvement.

c) Prediction Distribution

The model predicted the following number of samples for each class:

Class	Number of Predictions
Benign	47
Malignant	34
Normal	36

This distribution shows that the model predicted all three classes, although the predictions are not perfectly balanced.

6. CONCLUSION

The Hybrid CNN–Quantum model was successfully trained to classify breast ultrasound images into benign, malignant, and normal categories. With further improvements such as more training data, better feature extraction, or deeper networks, the classification performance can be further enhanced.